

C.4

Standing waves and resonance

Guiding questions

- What distinguishes standing waves from travelling waves?
- How does the form of standing waves depend on the boundary conditions?
- How can the application of force result in resonance within a system?

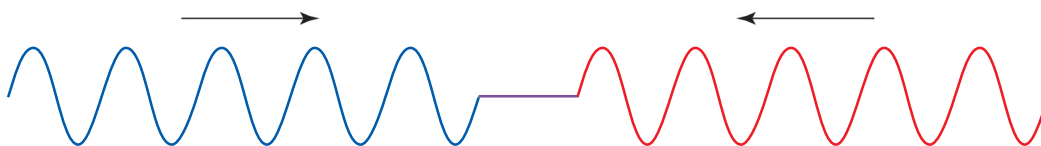
The nature of standing waves

SYLLABUS CONTENT

- ▶ The nature and formation of standing waves in terms of superposition of two identical waves travelling in opposite directions.
- ▶ Nodes and antinodes, relative amplitude and phase difference of points along a standing wave.

So far, the discussion of waves has been about travelling waves, which transfer energy progressively away from a source. Now we turn our attention to waves that remain in the same position.

Consider two travelling waves of the same shape, frequency, wavelength and amplitude moving in opposite directions, such as shown in Figure C4.1, which could represent transverse waves on a string or a rope.

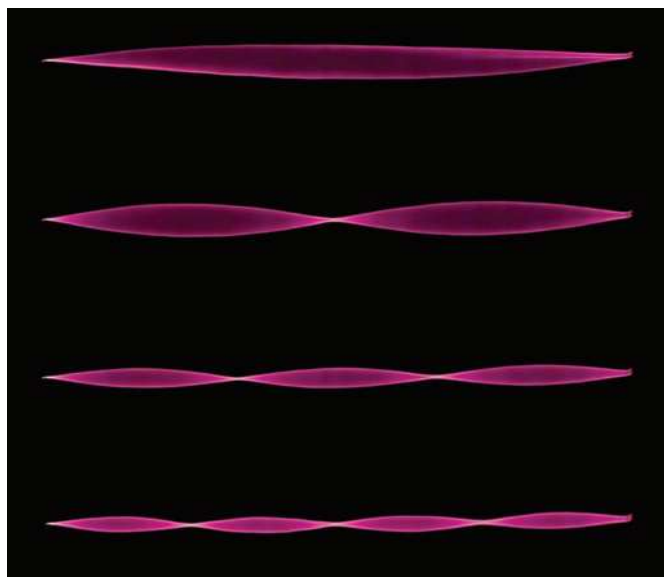


■ **Figure C4.1** Two travelling waves of the same shape, frequency, wavelength and amplitude moving in opposite directions

As these waves pass through each other, they will combine to produce an oscillating wave pattern that does not change its position. Such patterns are called standing waves.

Standing waves usually occur in confined systems in which waves are reflected back upon each other repeatedly.

Typical examples of standing wave patterns (on strings) are shown in Figure C4.2. Note that a camera produces an image over a short period of time (not an instantaneous image) and that is why the fast-moving string appears blurred. This is also true when we view such a string with our eyes.



■ **Figure C4.2** Standing waves on a stretched string

Simple standing wave patterns can be produced by shaking one end of a rope, or long stretched spring, at a suitable frequency, while someone holds the other end stationary. Patterns like those shown in Figure C4.2 require high frequencies (because a string is much less massive than a rope), but can be produced in a laboratory by vibrating a stretched string with a mechanical vibrator which is controlled by variable electrical oscillations from a signal generator. This apparatus can be used to investigate the places at which the string appears to be stationary and the frequencies at which they occur.

◆ **Standing wave** The kind of wave that can be formed by two similar travelling waves moving in opposite directions. The most important examples are formed when waves are reflected back upon themselves. The wave pattern does not move and the waves do not transfer energy.

◆ **Nodes** The positions in a standing wave where the amplitude is zero.

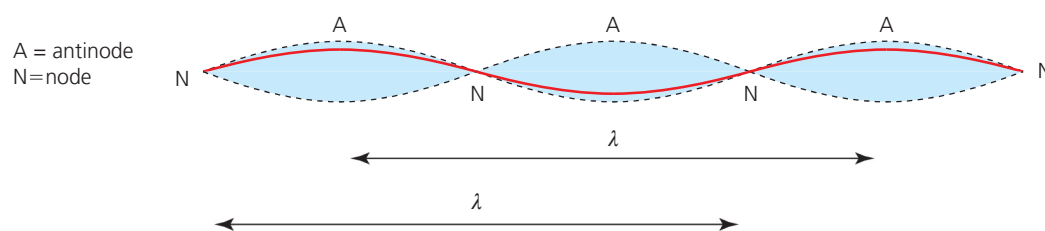
◆ **Antinodes** The positions in a standing wave where the amplitude is greatest.

Nodes and antinodes

A standing wave pattern remains in the same place.

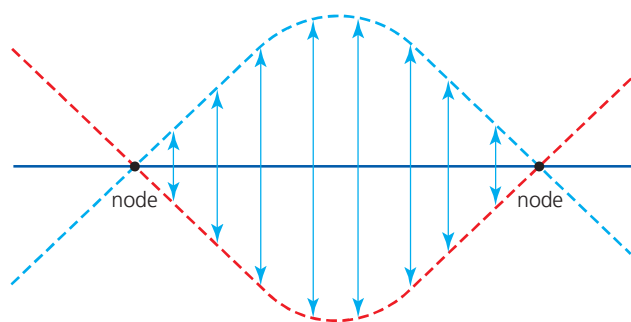
Points where the displacement is always a minimum (often zero) are called **nodes**. Points where the amplitude is greatest are called **antinodes**.

Figure C4.3 represents the third wave from the photograph in Figure C4.2 diagrammatically. Note that the distance between alternate nodes (or antinodes) is one wavelength.



■ **Figure C4.3** Nodes and antinodes in a standing wave

A system in which there is a standing mechanical wave has both kinetic energy and potential energy, but energy is not transferred *away* from the system in the form of a wave along the system. However, there will be energy dissipation within the system because of resistive forces, so that the amplitude of the standing wave will decrease, unless it is sustained by energy transferred from an external driving frequency (see later).



■ **Figure C4.4** Variation of amplitude between nodes

Nodes occur at places where the two waves are *always* exactly out of phase. At other places, the displacements will oscillate between zero and a maximum value which depends on the phase difference between the waves moving in opposite directions. At the antinodes the two waves are always exactly in phase. You should try using a computer simulation to illustrate this time-changing concept.

Between adjacent nodes all parts of the medium oscillate in phase with each other with the same frequency. Each position has a constant amplitude, but the amplitudes vary as shown in Figure C4.4.

Standing waves are possible with any kind of wave (transverse or longitudinal) moving in one, two or three dimensions. But, for simplicity, discussion will be confined to one-dimensional waves, such as transverse waves on a stretched string, or longitudinal sound waves in pipes.

■ Boundary conditions

Standing waves occur most commonly when waves are reflected repeatedly back from boundaries in a confined space. The frequencies of the standing waves will depend on the nature of the ends of the system. These are called the **boundary conditions**. For example, the ends of a string, or rope, may be fixed in one position or free to move; the ends of a pipe could be closed or open. When the ends are free to move, we would assume that standing wave has antinodes there. There will be nodes at fixed ends.

◆ Boundary conditions

The conditions at the ends of a standing wave system. These conditions affect whether there are nodes or antinodes at the ends.

The boundary conditions of a standing wave system describe whether there are nodes or antinodes at the end of the system.

Comparing standing waves with travelling waves

■ **Table C4.1** Comparison of standing waves and travelling waves

	Standing waves	Travelling waves
Wave pattern	standing (stationary)	travelling (progressive)
Energy transfer	no energy is transferred	energy is transferred through the medium
Amplitude (assuming no energy dissipation)	amplitude at any one place is constant but it varies with position between nodes: maximum amplitude at antinodes, zero amplitude at nodes	all oscillations have the same amplitude
Phase	all oscillations between adjacent nodes are in phase	oscillations one wavelength apart are in phase; oscillations between are not in phase
Frequency	all oscillations have the same frequency	all oscillations have the same frequency
Wavelength	twice the distance between adjacent nodes	shortest distance between points in phase

■ Standing wave patterns in strings

SYLLABUS CONTENT

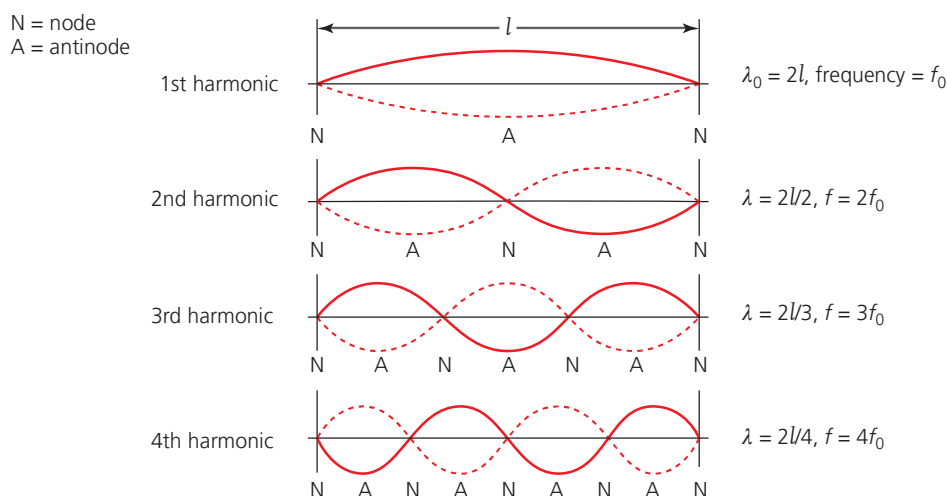
- ▶ Standing waves patterns in strings.

◆ **Harmonics** Different frequencies (modes) of standing wave vibrations of a given system. The frequencies are all multiples of the frequency of the **first harmonic**.

◆ **Modes of vibration** The different ways in which a standing wave can arise in a given system.

If a stretched string *fixed at both ends* is struck or plucked, it can only vibrate as a standing wave with nodes at both ends. The amplitude of the standing wave will usually decrease quickly as energy is dissipated.

The simplest way in which it can vibrate is shown at the top of Figure C4.5. This is known as the **first harmonic**. It is usually the most important mode of vibration, but a series of other harmonics (**modes of vibration**) is possible and can occur at the same time as the first harmonic. Some of these harmonics are shown in Figure C4.5.



■ **Figure C4.5** Modes of vibration of a stretched string fixed at both ends

The first harmonic is the standing wave with the lowest possible frequency, f_0 (greatest wavelength). Other harmonics are mathematical multiples of this frequency.

If a string is fixed at both ends, the wavelength, λ_0 , of the first harmonic is $2l$, where l is the length of the string. The speed of the wave, v , along the string depends on the tension and the mass per unit length. If the wave speed is known, the frequency of the first harmonic, f_0 , can be calculated using $v = f\lambda$ (from Topic C.2):

$$f_0 = \frac{v}{\lambda_0} = \frac{v}{2l}$$

The wavelengths of the harmonics are, starting with the first (longest), $2l$, $\frac{2l}{2}$, $\frac{2l}{3}$, $\frac{2l}{4}$ and so on. The corresponding frequencies, starting with the lowest, are f_0 , $2f_0$, $3f_0$, $4f_0$ and so on.

The wavelength of the first harmonic can be found from the length of the system and the boundary conditions.

$v = f\lambda$ can then be used to connect frequency and wave speed.

WORKED EXAMPLE C4.1

A stretched rope with two fixed ends has a length of 0.98 m, and waves travel along it with a speed of 6.7 ms^{-1} .

a Calculate:

- the wavelength of the first harmonic
- the frequency of the first harmonic.

b Calculate:

- the wavelength of the fourth harmonic
- the frequency of the fourth harmonic.

c Explain how your answers would change if the tension in the rope was increased.

Answer

a i $\lambda_0 = 2l = 2 \times 0.98 = 2.0 \text{ m}$ (1.96 seen on calculator display)

ii $f_0 = \frac{v}{\lambda_0} = \frac{6.7}{1.96} = 3.4 \text{ Hz}$

b i $\lambda = \frac{2l}{4} = \frac{\lambda_0}{4} = \frac{1.96}{4} = 0.49 \text{ m}$

ii $f = 4f_0 = 4 \times 3.4 = 14 \text{ Hz}$

c The wavelengths would remain the same, but the frequencies would increase because the wave would travel faster if the tension was increased.

Standing waves on strings and ropes are usually between fixed ends, but it is possible that one, or both, boundaries could be 'free'. If the two ends are free (an unusual event), there will be antinodes at each end / boundary, so that the frequency of the first harmonic will be the same as for fixed boundaries, which has nodes at each end.

◆ **Oscilloscope** An instrument for displaying and measuring potential differences that change with time.

◆ **Waveform** Shape of a wave.

If there is a node at one end and an antinode at the other, the wavelength of the first harmonic will be greater and its frequency will be lower. An example of this situation would be a standing wave produced on a chain hanging vertically. (In that case, the wavelength would not be constant: see Question 4.)

Inquiry 1: Exploring and designing

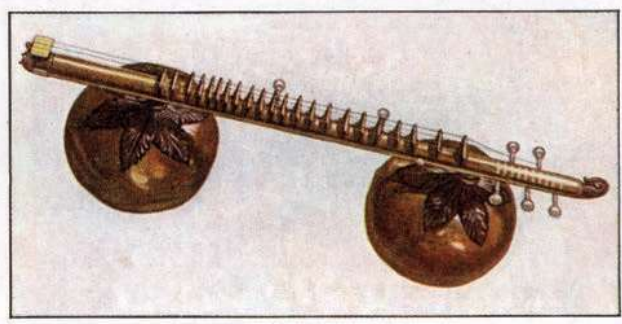


Designing

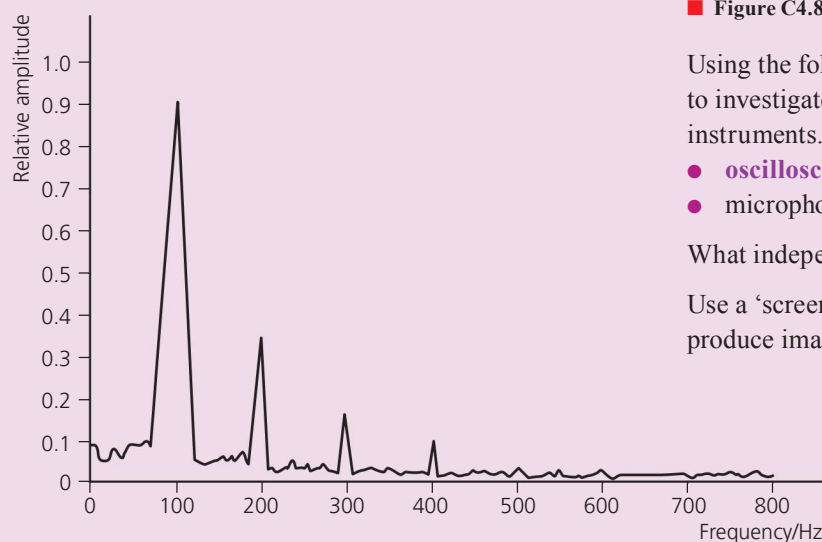
Musical instruments

An amazing variety of musical instruments have been used all around the world for thousands of years (see Figure C4.6). Most involve the creation of standing wave patterns (of different frequencies) on strings, wires, surfaces or in tubes of some kind. The vibrations disturb the air around them and thereby send out sound waves (musical notes).

When musical notes are played on stringed instruments, such as guitars, cellos and pianos, the strings vibrate mainly in their first-harmonic modes, but various other harmonics will also be present. This is one reason why each instrument has its own, unique sound. Figure C4.7 shows a range of frequencies that might be obtained from a guitar string vibrating with a first harmonic of 100 Hz.



■ **Figure C4.6** Vina, an Indian stringed instrument



■ **Figure C4.7** Frequency spectrum from a guitar string

The factors affecting the frequency of the first harmonic are the length of the string, the tension and the mass per unit length. For example, middle C has a frequency of 261.6 Hz. The standing transverse waves of the vibrating strings are used to make the rest of the musical instrument oscillate at the same frequency. When the vibrating surfaces strike the surrounding air, travelling longitudinal sound waves propagate away from the instrument to our ears.



■ **Figure C4.8** Creating standing waves on a cello

Using the following apparatus, design an experiment to investigate the sound waves produced by different instruments.

- **oscilloscope** or app on computer or phone
- microphone.

What independent variables will you change or control?

Use a 'screen capture' technique on the oscilloscope to produce images and compare the **waveforms** produced.

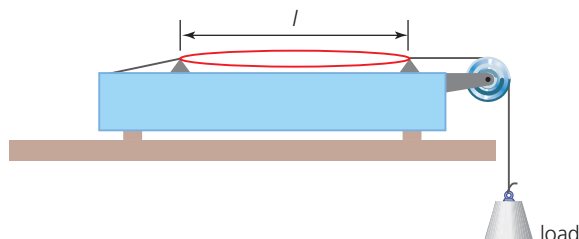
WORKED EXAMPLE C4.2

An experiment similar to that shown in Figure C4.9 can be used to determine the speed of a wave along a stretched string.

- If the length of the vibrating string was 79.4 cm and the first harmonic had a frequency of 140 Hz, calculate the wave speed.
- Determine the wavelength and frequency of the fifth harmonic.
- A student finds the following formula on the internet:

$$\text{wave speed} = \sqrt{\left(\text{tension} \times \frac{\text{length}}{\text{mass}} \right)}$$

If the mass of the vibrating string was 2.6 g, and the tension was 147 N, show that use of this formula confirms the answer to part a.



■ **Figure C4.9** Experiment to determine wave speed

Answer

$$\text{a } v = f_0 \lambda_0 = 140 \times (2 \times 0.794) = 222 \text{ m s}^{-1}$$

$$\text{b } 5f_0 = 5 \times 140 = 700 \text{ Hz}$$

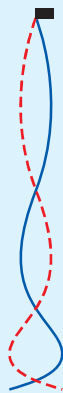
$$\frac{\lambda_0}{5} = \frac{1.588}{5} = 0.318 \text{ m}$$

$$\begin{aligned} \text{c } v &= \sqrt{\left(147 \times \frac{0.794}{0.0026} \right)} \\ &= 2.1 \times 10^2 \text{ m s}^{-1} \end{aligned}$$

This is within 5% of the answer to part a, so the two answers are consistent, within experimental uncertainties.

- A string on a violin has a length of 32.8 cm and produces a note of 262 Hz (middle C).
 - Calculate the speed of the wave on the string.
 - State any assumption that you made.
 - Suggest what will happen to the speed of the wave on the string and the frequency of the note if sometime later the same string has lost some tension.
- The velocity of a transverse wave on a string of length 28 cm is 240 m s^{-1} .
 - Calculate the frequency of the second harmonic of a standing wave on this string when both ends are fixed.
 - Determine the wavelength of the sound that this produces in the surrounding air. Assume the speed of sound in air is 340 m s^{-1} .
- A teacher wishes to show his class standing waves on a thin string, similar to those seen in Figure C4.2. He uses a vibration generator set at a frequency of 384 Hz.
 - Determine the length of string that is needed to produce the fourth harmonic if the speed of the wave is 285 m s^{-1} .
 - Using the same apparatus, the teacher increases the length of the string to exactly 2.00 m. Predict the wavelength and frequency of the third harmonic in this new arrangement.

- 4 The top of a thin chain hanging vertically is shaken with increasing frequency until the standing wave pattern seen in Figure C4.10 is produced.
- Suggest why the wavelength on this system decreases towards the lower end of the chain.
 - Describe the boundary conditions of this system.
 - State which harmonic is shown in the picture.
 - Explain why it is not possible to produce the second harmonic on this system.

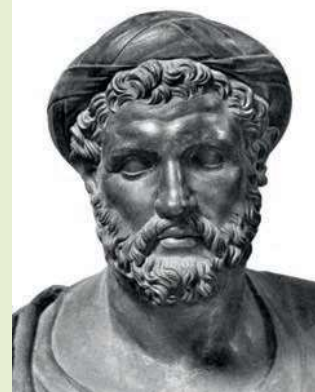


■ Figure C4.10 A standing wave on a hanging chain.



Nature of science: Patterns and trends

Pythagoras is often credited with being the first to realize that there was a mathematical relationship between musical notes and the dimensions of the instrument that produced them. That was about 2500 years ago. More generally, this may have been one of the first occasions when mathematical / 'scientific' reasoning was used to describe features of everyday life.



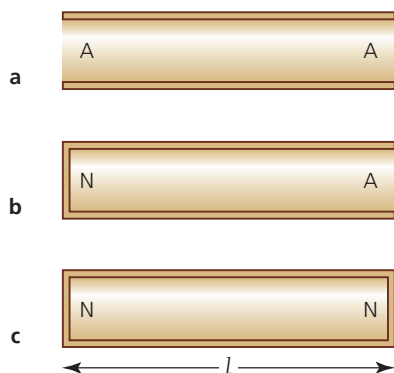
■ Figure C4.11 Pythagoras

Standing wave patterns of air in pipes

SYLLABUS CONTENT

- Standing waves patterns in pipes.

Standing longitudinal waves of sound can be created easily in the air contained by a pipe / tube / column. The air may be set into motion by, for example, the simple action of blowing across the open end of the pipe. The sound produced by blowing across the top of an empty bottle is an everyday example. Musical wind instruments, like trumpets or flutes, use the same principle.



■ Figure C4.12 Nodes and antinodes at the ends of open and closed pipes

As with strings, in order to understand which wavelengths can be produced, we need to consider the length of the system and the boundary conditions.

The first harmonic is the standing wave with the greatest possible wavelength, λ_0 , and lowest possible frequency, f_0 . Other harmonics are multiples of this frequency.

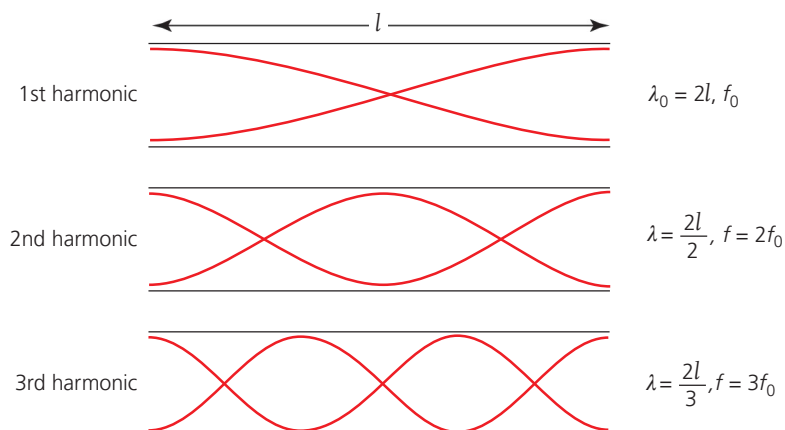
After determining possible wavelengths, the equation $v = f\lambda$ can then be used to predict harmonic frequencies if the wave speed is known.

Figure C4.12 show the three possible combinations of boundary conditions.

A pipe open at both ends must have antinodes, A, at the ends, and at least one node, N, in between. A pipe open at only one end has one antinode and one node as its boundary conditions.

A pipe closed at both ends (an unusual situation) must have nodes at the ends and at least one antinode in between.

Figure C4.13 graphically represents the first three harmonics (vibration modes) for a pipe open at both ends.

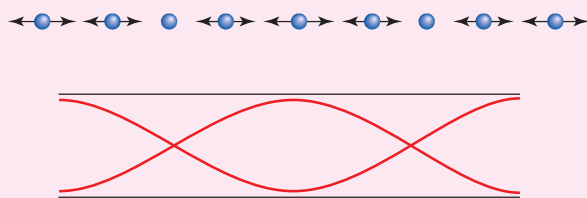


■ **Figure C4.13** The first three harmonics in a pipe open at both ends

The wavelength of the first harmonic (twice the distance between adjacent nodes or antinodes) is $2l$.

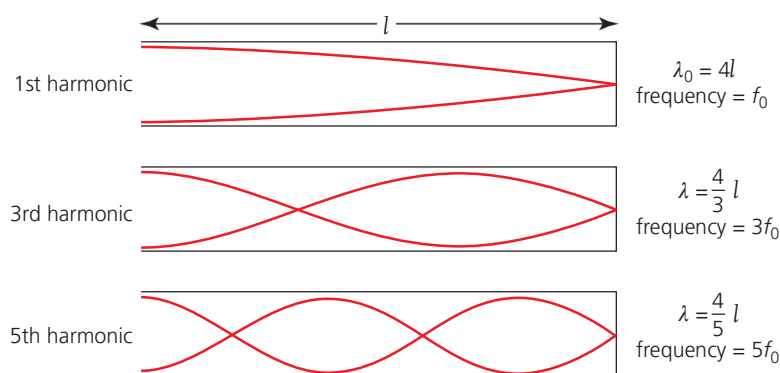
Common mistake

Note that the representations of standing longitudinal waves seen in Figure C4.13 and Figure C4.15 may cause confusion: the curved lines in the diagrams are an indication of the maximum *sideways* displacement of vibrating air molecules. They should not be mistaken for transverse waves, like those on a string. Figure C4.14 shows how the second harmonic shown in Figure C4.13 is representing the movements of some molecules.



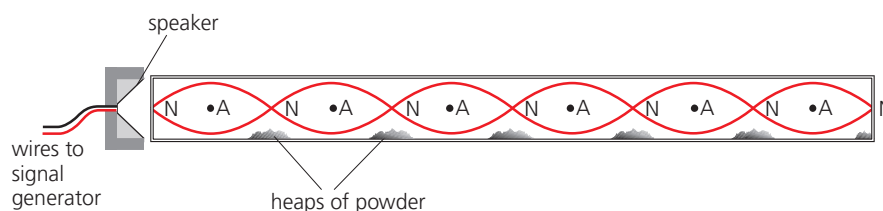
■ **Figure C4.14** How the second harmonic can represent the movements of some molecules

For a pipe which is closed at one end, but open at the other, the possible standing waves are shown in Figure C4.15. The first harmonic has a longer wavelength and lower frequency than for a similar length pipe which is closed, or open, at both ends. Note that only odd-numbered harmonics are possible with these boundary conditions.



■ **Figure C4.15**
Harmonics in a pipe
open at one end

One way of demonstrating standing sound waves in air is by using a small loudspeaker attached to the end of a long, clear plastic tube, which is closed at the other end. See Figure C4.16. Electrical signals of different frequencies can be applied to the loudspeaker which then sends sound waves of the same frequencies down the tube. When the waves reflect back off the end of the tube, a standing wave can be set up only *if* the frequency is equal to the frequency of the one of the possible harmonics. Some powder can be scattered all along the pipe and when the loudspeaker is turned on and the frequency carefully adjusted, the powder is seen to move into separate piles. This is because the powder tends to move from places where the vibrations of the air are large (antinodes) to the nodes, where there are no vibrations. The tube may be considered as closed at both ends.



■ **Figure C4.16**
Demonstrating a standing wave with a loudspeaker

WORKED EXAMPLE C4.3

Consider Figure C4.16.

- State which harmonic is shown in Figure C4.16.
- Calculate the wavelength of this standing wave if the length of the tube is 73.5 cm.
- Determine the speed of the sound wave in air if the frequency used by the loudspeaker was 1410 Hz.
- Calculate the theoretical frequencies of the first two observable harmonics if the tube was left open at the end on the right-hand side.

Answer

- sixth
- $73.5/3 = 24.5$ cm
- $v = f\lambda = 1410 \times 0.245 = 345 \text{ m s}^{-1}$ (345.45 seen on calculator display)
- See Figure C4.15. The wavelength of the first harmonic will be $4 \times$ length of tube = 2.94 m.

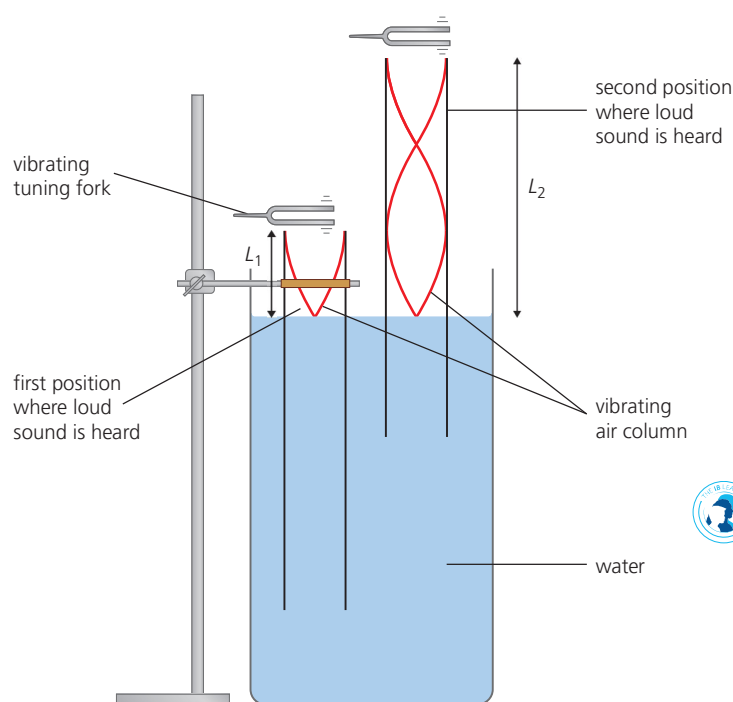
$$f_0 = \frac{v}{\lambda} = \frac{345.45}{2.94} = 1.18 \times 10^2 \text{ Hz (117.5 seen on calculator display)}$$

The second harmonic does not occur.

$$\text{Frequency of third harmonic} = 3f_0 = 3 \times 117.5 = 353 \text{ Hz}$$

Figure C4.17 shows another way of investigating standing waves in pipes. The length of the pipe is easily changed by raising or lowering it in the tall container of water. The pipe is always open at one end and closed at the other. Air in the pipe is disturbed by the pure, single frequency emitted from a **tuning fork** (see Figure C4.20) placed just above the open end of the tube, and the length of the pipe is adjusted until the sound of the standing wave becomes audible.

◆ **Tuning fork** Device designed to vibrate at only one precise frequency.



■ **Figure C4.17** Demonstrating standing waves with a tuning fork

The length of the tube seen on the left of the Figure C4.17 has been adjusted to the shortest length for which a standing wave can be heard. This will be the first harmonic for a tube of that length. The position on the right corresponds to the third harmonic for a tube of the new length. (The speed, wavelength and frequency of the wave are unchanged.) There is no second harmonic possible with these boundary conditions (open at one end, closed at the other). Higher harmonics may be heard if the tube and container are long enough.

ATL C4A: Social skills

Working collaboratively

Suggest how physics and music students could work together to produce a presentation called 'The Science of Music' for the rest of their year group.

Consider how you could share your presentations online for future students.

For these questions, where necessary, assume that the speed of sound in air is 342 m s^{-1} .

- 5 a If the tuning fork in Figure C4.17 had a frequency of 384 Hz , show that the length of the pipe needed for the first harmonic to be heard is about 20 cm .
b How far will the pipe need to be raised until the next harmonic is heard?
- 6 Sketch the first three harmonics for air in a pipe which is closed at both ends.
- 7 A pipe has a length of 1.32 m and is closed at both ends. Determine the wavelength and frequency of its third harmonic.
- 8 The flute is the oldest of musical instruments. See Figure C4.18. There are a very large number of designs. It can be considered as a pipe which is open at both ends. Sound is produced when air is blown across an opening near the end of the pipe.
 - a Determine what length (cm) of pipe will produce a first harmonic of frequency 493 Hz .
 - b Explain the purpose of the holes along the length of the flute.
 - c A clarinet is a similar type of musical instrument to a flute, but the pipe is closed at one end. Compare the length of the pipes of a flute and clarinet that produce musical notes of the same frequency.



■ **Figure C4.18** Radha listening to Krishna's flute

- 9 Apparatus similar to that shown in Figure C4.17 was used to investigate the variation of the speed of sound with air temperature. At a temperature of 30°C , adjacent nodes were found to be separated by a distance of 23.3 cm when using a frequency of 750 Hz .
 - a Determine the speed of sound at this temperature.
 - b Give a molecular explanation of why this speed is greater than the 342 m s^{-1} used in other questions.
- 10 Water being poured into a bottle may produce many sounds, but there will usually be a noticeable increase in frequency of the sound as the bottle fills up. Explain this effect.

The natural sciences

- What is the role of imagination and intuition in the creation of hypotheses in the natural sciences?

When electrons were discovered in 1897, they were believed to be tiny solid particles. Twenty-seven years later, it was proved that electrons have wave properties (see Theme E). This led on to the theory that electrons could only be confined within atoms in the form of standing waves.

Most people will struggle to visualize electrons as three-dimensional standing waves in unimaginably small atoms. But, after about one hundred years, this still remains the accepted theory. There is no good reason for us to believe that the atomic-scale particles must behave in ways that are the same as masses which we can see with our eyes.

Natural frequencies of vibration

SYLLABUS CONTENT

- The nature of natural frequency.
- The effects of light, critical and heavy damping on the system.

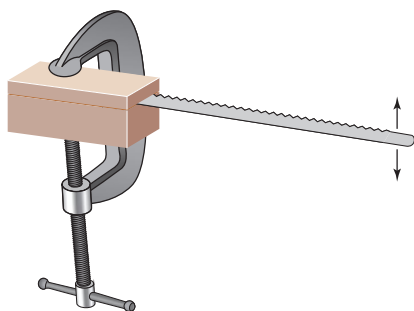
◆ Frequency, natural

The frequency at which a system oscillates when it is disturbed and then left to oscillate on its own, without influence from outside.

◆ **Vibration** Mechanical oscillation (usually of relatively small amplitude).

When many objects are struck briefly by an external force, they vibrate ‘freely’, or ‘naturally’ (although, for most objects, the vibrations may be insignificant and/or very short-lasting because energy is quickly dissipated into the surroundings). Such vibrations often disturb the air around them and send longitudinal waves into the environment, which may be heard as sound, if they have a suitable frequency.

When an object is disturbed and then left to oscillate without further interference, it is said to oscillate at its **natural frequency** (or frequencies).



■ **Figure C4.19** Vibrating hacksaw blade

Relatively small amplitude *mechanical* oscillations are commonly called **vibrations**. The simplest examples of natural frequencies are those of a simple pendulum and mass on a spring, as discussed in Topic C.1. A further example is a clamped ruler, or hacksaw blade, as shown in Figure C4.19.

Vibrating objects will oscillate at a natural frequency(s) which depends on their dimensions and masses.

Tool 3: Mathematics

Linearize a graph

A student has read that the square of the time period of a hacksaw blade oscillator is proportional to the cube of the vibrating length.

What graph, involving frequency, should she draw in order to see if the relationship is correct?

An object made of only one material in a simple shape, such as a tuning fork (see Figure C4.20), may produce a single, ‘pure’, natural frequency. But most objects will have more complicated structures and a range of natural frequencies, although one frequency may dominate.

The *two-dimensional* standing wave patterns of a horizontal metal plate can be observed by placing small grains, such as fine sand or salt, onto a surface that is disturbed into vibration at the plate's natural frequencies. See Figure C4.21. (Oscillations can be easily maintained using a mechanical oscillator, driven by a signal generator, to vibrate the plate.)



■ **Figure C4.20** This tuning fork produces a frequency of exactly 440 Hz (a musical A)



■ **Figure C4.21** Demonstrating the oscillations of a metal plate

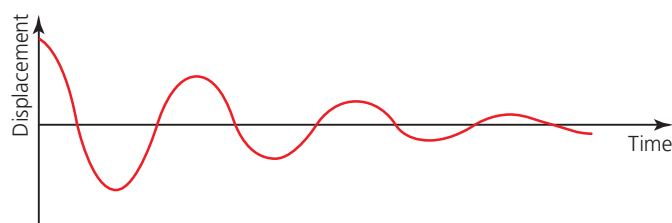
Damping

The motions of all macroscopic objects have resistive forces of one kind or another acting against them. Resistive forces will always act in the opposite direction to the instantaneous motion of an oscillator, and result in a reduction of its speed and kinetic energy.

Therefore, as with all other mechanical systems, useful energy is transferred from an oscillator into the surroundings (dissipated) in the form of thermal energy and maybe some sound. Consequently, an oscillator will move at slower and slower speeds, and its successive amplitudes will decrease in size. This effect is called damping.

Damping is the dissipation of energy from an oscillator due to resistive forces.

It is common for the frequency (and time period) of a vibration to remain approximately constant during damping, as shown by the graph in Figure C4.22. This is because, although the displacements are reduced, the speeds and accelerations also decrease.



■ **Figure C4.22** Decreasing amplitude (at constant frequency) of a damped oscillation

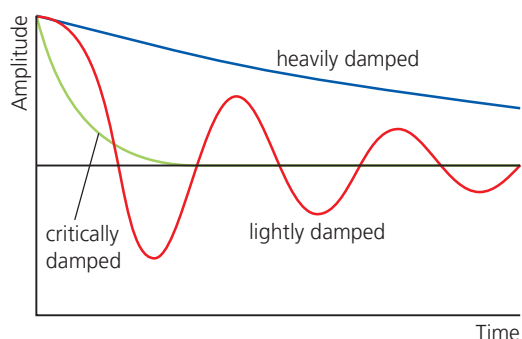
Top tip!

The magnitude of each successive peak of the graph shown in Figure C4.22 can be determined by multiplying the previous value by the same fraction. For example: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ (This is known as an exponential series, or a geometric sequence.)

Tool 3: Mathematics

Propagate uncertainties in processed data

Successive amplitudes (in cm) of the **free vibration** of a hacksaw blade were measured to be 2.7, 2.4, 2.1, 1.8, 1.6, 1.4, 1.3, 1.1. The measurements were made to the nearest millimetre. Considering uncertainties in measurement, is it possible that this data fits an exponential pattern?



■ **Figure C4.23** Light damping, heavy damping and critical damping

◆ **Vibration (free)**

Vibration without any external influence.

◆ **Damping (critical)**

When an oscillating system returns relatively quickly to its equilibrium position without oscillating.



■ **Figure C4.24** A car's shock absorber

Damping can be investigated experimentally using simple apparatus like that shown in Figure C4.19, but with horizontal cards of different areas taped to the blade in order to increase air resistance.

The amount (degree) of damping in oscillating systems can be very different, as shown in Figure C4.23.

Some oscillations are *heavily damped* because of considerable frictional forces. In effect no oscillations occur because resistive forces are such that the object takes a long time (compared to its natural time period) to return to its equilibrium position.

Conversely, an oscillator may be *lightly damped*, so that it continues to oscillate, taking a relatively long time to dissipate its energy. A pendulum and a mass oscillating on a spring are good examples of lightly damped systems.

If an oscillation is opposed by resistive forces, such that it settles relatively quickly (compared to its natural time period) back into its equilibrium position, without ever passing through it, the process is described as **critical damping**. A car's suspension (see Figure C4.24) is an example of this kind of damping.

11 Describe how the natural frequency of

- a a simple pendulum,
- b a mass hanging vertically from a spring can be increased.

12 A student was investigating the vibrations of a hacksaw blade, as shown in Figure C4.19. She displaced the end of the blade and then left it to vibrate freely, but she found that the vibrations were too quick for her to observe easily.

- a Suggest how she could decrease the frequency of the vibrations.
- b The blade then vibrated with a frequency of 1.0 Hz, while its amplitude was (almost) constant, at 0.50 cm, for several seconds. Sketch a displacement–time graph for the first two seconds of its motion.
- c The vibrations were then damped. Add a second line to your sketch for part b, to represent the damped oscillations.
- d Suggest how the student could have damped the vibrations.

13 Outline why the fine sand shown in Figure C4.21 moves into places which demonstrate the standing wave pattern on the plate.

14 Figure C4.25 shows an automatic door closer.

- a Describe its intended purpose.
- b What type of damping does it exhibit?
- c Give an example of where it might be used.



■ **Figure C4.25** Automatic door closer

◆ Vibration (forced)

Vibration affected by external periodic forces.

■ Forced vibrations

A **forced vibration** (oscillation) occurs when an external oscillating (periodic) force acts on a system. This may tend to make it oscillate at a frequency which is different from its natural frequency.



■ **Figure C4.26** How can we increase the amplitude of a swing?

We are surrounded by a range of oscillations. It is important to consider how these oscillations affect other things around them. In other words, what happens when an external oscillating force is continuously applied to a separate system?

To understand this, it is helpful to consider a very simple example: what happens when we keep pushing a child on a swing (see Figure C4.26)?

In this case it is fairly obvious: it depends on when we push the swing and in which direction. If we want bigger swings (increasing amplitudes), then we should push once every oscillation in the direction in which the child is moving at that moment. In more scientific terms, we would say that we need to apply an external force that has the same frequency as, and is in phase with, the natural frequency of the swing.

The most important examples of forced oscillations are those in which the frequency of the external force (**driving frequency**) is the same as the natural frequency. The child on the swing described above is a good example of this. When a regular periodic stimulus to a system results in an increasing amplitude the effect is called resonance.

◆ **Frequency, driving** The frequency of an oscillating force (periodic stimulus) acting on a system from outside. Sometimes called forcing frequency.

◆ **Resonance** The increase in amplitude that occurs when an oscillating system is acted on by an external periodic force that has the same frequency as the natural frequency of the system. The driving force must be in phase with the natural oscillations of the system.

◆ **Frequency–response graph** Graph used to show how the amplitude of a system's oscillations responds to different driving frequencies.

◆ **Resonant frequency** The frequency at which resonance occurs.

■ Resonance

SYLLABUS CONTENT

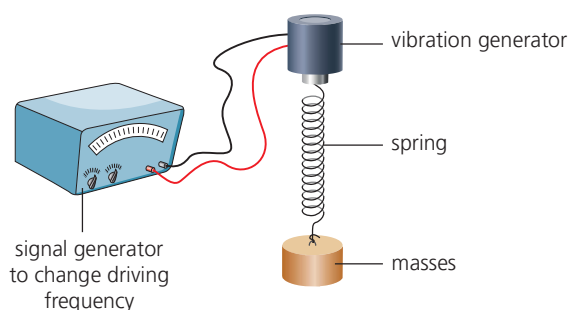
- ▶ The nature of resonance including the amplitude of oscillations based on driver frequency.
- ▶ The effect of damping on the maximum amplitude and resonant frequency of oscillation.

Resonance is the name given to the increase in amplitude and energy of an oscillation that occurs when a periodic external driving force has the same frequency as the natural frequency of a system.

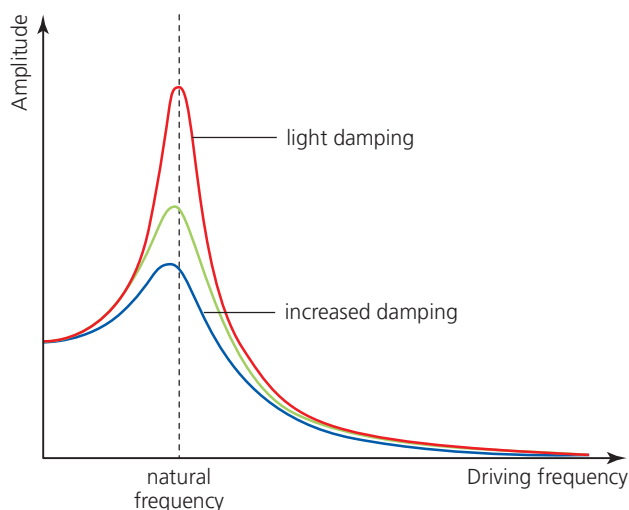
The oscillations of the driving force must be in phase with the natural oscillations of the system. Simple quantitative laboratory experiments into the effects of resonance can be difficult to perform, but they can produce interesting results that show how varying the driving frequency affects the amplitude of an oscillating system. When the force is first applied, the oscillations may be erratic, but the system will settle into a regular pattern of movement with a measurable maximum amplitude.

Figure C4.27 shows a possible arrangement. The resonant frequency of the mass–spring system can be changed by using different springs and/or different masses. The driving frequency is provided by the vibration generator, which can be driven using different frequencies from the signal generator.

A typical **frequency–response graph** drawn from the results of an experiment like that shown in Figure C4.28 rises to a maximum amplitude at the **resonant frequency**. This occurs when the rate of energy dissipation (damping) has risen to a level that is equal to the power supplied from the source of the driving frequency.



■ **Figure C4.27** Investigating the resonance of a mass on a spring



■ **Figure C4.28** Typical frequency–response curves with different degrees of damping

The resonant frequency is at, or very close to, the natural frequency, but the sharpness and height of the peak also depend on the amount of damping in the system. The greater the damping, the greater the dissipation of energy and, therefore, the smaller the amplitude. The value of the resonant frequency reduces slightly with greater damping.

If there is a powerful input, or very little damping, amplitudes of vibration can become large and this may have destructive, or useful, consequences.

The energy of an oscillation is proportional to its amplitude *squared*.

There may be smaller resonance peaks at values of the driving frequency which are equal to the natural / resonant frequency divided by 2, 3, 4 and so on (not shown in the diagram).

LINKING QUESTIONS

- How does the amplitude of vibration at resonance depend on the dissipation of energy in the driven system?
- How can resonance be explained in terms of conservation of energy?

These questions link to understandings in Topic A.3.

Inquiry 2: Collecting and processing data

Processing data

Table C4.2 shows the results obtained in an experiment similar to that seen in Figure C4.27. The uncertainties in this experiment were significant.

Draw a graph of these results, including uncertainty bars, with a curve of best fit to determine a value for the resonant frequency. You should not assume that the largest recorded measurement is the peak of the graph.

■ Table C4.2 Results of resonance experiment

Applied frequency / Hz ± 2 Hz	Maximum amplitude / cm ± 0.5 cm
4	1.7
6	2.1
8	2.7
10	4.9
12	5.4
14	3.3
16	2.5
18	1.9

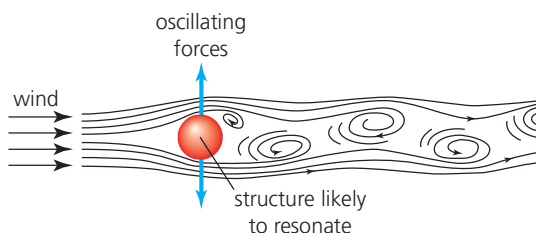


Examples of resonance

Some examples of resonance are useful but many more are unwanted, and we usually try to reduce their destructive effects. Avoiding resonance in all types of structure is a major concern for engineers and it provides an interesting combination of physics theory and practical engineering.



■ **Figure C4.29** Resonance may be one reason why some buildings collapse in an earthquake



■ **Figure C4.30** A steady wind can cause oscillating forces because of the alternate way in which vortices can be formed



■ **Figure C4.31** Walker on a suspension bridge in Nepal



■ **Figure C4.32** The Millennium Bridge in London was affected by resonance

Unwanted resonance

Parts of almost all machinery (and their immediate surroundings) may vibrate destructively when their motors are operating at certain frequencies. For example, a washing machine may vibrate strongly when the spinner is running at a certain frequency, and parts of vehicles can vibrate when the engine reaches a certain frequency, or they travel at certain speeds.

Earthquakes may well affect some buildings more than others. The buildings that are most damaged are often those that have natural frequencies closest to the frequencies of the waves produced by the earthquake (Figure C4.29).

Strong but steady winds, or currents, can also cause dangerous resonance in structures such as bridges and towers. This is often due to the effect of eddies and vortices as the wind or water flows around the structures. See Figure C4.30.

If you have ever crossed a small suspension bridge for walkers (such as the one in Figure C4.31), you will probably know how easy it is to set it vibrating with increasing amplitude by shaking it or stamping your feet at a certain frequency. This is because it would be too difficult or expensive to build such a simple bridge with a natural frequency that is very different from a frequency that people can easily reproduce, or to use a design that incorporated damping features.

The resonance of bridges has been well understood for many years and the flexibility of suspension bridges makes them particularly vulnerable. The famous collapse of the newly built Tacoma Narrows Bridge in the USA in 1940 is widely given as a simple example of resonance caused by the wind, although this is only part of a much more complex explanation. Videos of the collapse are easily found on the internet. In June 2000, the Millennium Bridge across the River Thames in London had to be closed soon after its opening because of excessive lateral (sideways) oscillations due to resonance (Figure C4.32).

In this case positive feedback was important. The slow oscillations of the bridge made people sway with the same frequency, and their motion simply increased the periodic forces on the bridge that were causing resonance. The problem was solved by adding energy-dissipating dampers, but it was about 18 months before the bridge could reopen.

To reduce the risk of damage from resonance engineers can:

- alter the shape of the structure to change the flow of the air or water past it
- change the design so that the natural frequencies are not the same as any possible driving frequencies – this will involve changing the stiffness and mass of the relevant parts of the structure
- ensure that there is enough damping in the structure and that it is not too rigid, so that energy can be dissipated.

Useful resonance

- The molecules of certain gases in the atmosphere oscillate at the same frequency as thermal radiation emitted from the Earth. These gases absorb energy because of resonance; this results in the planet being warmer than it would be without the gases in the atmosphere. This is known as the greenhouse effect, as discussed in Topic B.2.
- Microwave ovens use electromagnetic radiation in the microwave region. The microwave wavelength equals a vibrational frequency of water molecules, so that the molecules absorb the radiation.
- Your legs can be thought of as pendulums with their own natural frequency. If you walk with your legs moving at that frequency, energy will be transferred more efficiently and it will be less tiring (we tend to do this without thinking about it).
- Quartz crystals can be made to resonate using electronics – the resulting oscillations are useful in driving accurate timing devices such as watches and computers.
- The sound from musical instruments can be amplified if the vibrations are passed on to a supporting structure that can resonate at the same frequency. An obvious example would be the strings on a guitar causing resonance in the box on which they are mounted. Because the box has a much larger surface area it produces a much louder sound than the string alone.
- Magnetic resonance imaging (MRI) is a widely used technique for obtaining images of features inside the human body. Electromagnetic waves of the right frequency (radio waves) are used to change the spin of protons (hydrogen nuclei) in water molecules in the patient's body.

LINKING QUESTION

- How can the idea of resonance of gas molecules be used to model the greenhouse effect? (NOS)

This question links to understandings in Topic B.2.

15 Consider the oscillations of a mass on a spring, as shown in Figure C1.5.

- Use the formula from Topic C.1 to determine the natural frequency of a 740 g mass oscillating vertically on a spring which has a spring constant of 17.2 N m^{-1} .
- Sketch a frequency–response curve to show how the amplitude of oscillation could change when the applied frequency from the vibration generator increases from 0.5 Hz to 2.0 Hz.
- Discuss how the results of the experiment will change if the mass was placed in water (in a beaker). Illustrate your answer by adding a second curve to your sketch for part **b** of this question.

16 a Estimate the natural frequency of your leg when it swings freely like a pendulum.

- If, when walking, your leg moves with the same frequency, predict your approximate speed, in km h^{-1} .

17 a Use the internet to find out a typical frequency of waves generated by an earthquake.

- Suggest ways in which an architect / civil engineer can ensure that their designs do not resonate dangerously at that frequency.

18 It is claimed that an opera singer can shatter a wine glass using sound resonance. Research the internet for any video evidence of this effect. Quote your conclusions and sources.

19 A wing mirror on a car resonates at multiples of its natural frequency of 20 Hz.

- Sketch a graph to show the frequency response of the mirror as the rpm (revolutions per minute) of the car engine increase from 1000 to 4000.
- Suggest how the vibrations of the mirror could be reduced.
- Add a second curve to your graph to show the new frequency response.

20 Carbon dioxide gas in the Earth's atmosphere, CO_2 , is an important cause of the greenhouse effect. In a simplified model, the molecule may be visualized as a simple harmonic oscillator, with the two oxygen atoms oscillating to and from a central carbon atom, as shown in Figure C4.33.

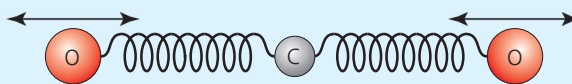


Figure C4.33 Oscillations of a carbon dioxide molecule

- If each oxygen molecule has a mass of $2.7 \times 10^{-27} \text{ kg}$, use the formula for the time period of a mass on a spring, with $k = 530 \text{ N m}^{-1}$, to determine a value for a resonant frequency of the carbon dioxide molecule.
- In what part of the electromagnetic spectrum are waves of this frequency to be found?